Windows Deployment Services (WDS)

Internal IT Knowledge Base / Standard Operating Procedure (SOP)

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, Help Desk, and Non-IT Staff
Environment	Windows Server
Domain	lab.local
Primary Service	WDS01 - Windows Deployment Services (WDS)
Related Services	DC01- Active Directory Domain Services (AD DS), DNS and DHCP01

1. Purpose

This document provides a standardized procedure for installing, configuring, testing, and maintaining Windows Deployment Services (WDS) in an organization's Windows environment.

WDS allows IT administrators to deploy Windows operating systems to computers over the network using PXE (Preboot Execution Environment).

Instead of manually installing Windows using a USB drive on every computer, IT can:

1. Connect the computer to the network.
2. Boot the computer using PXE.
3. Obtain an IP address from DHCP.
4. Contact the WDS server.
5. Download the Windows boot environment.
6. Select an approved Windows image.
7. Install Windows automatically or semi-automatically.

2. Scope

This procedure applies to:

- IT Administrators
- System Administrators
- IT Support personnel
- Desktop Support personnel
- Authorized technicians performing workstation deployment

This document covers:

- WDS installation
- WDS configuration
- DHCP integration
- PXE boot
- Boot image configuration
- Install image configuration
- WDS testing
- Basic troubleshooting
- Golden-image deployment

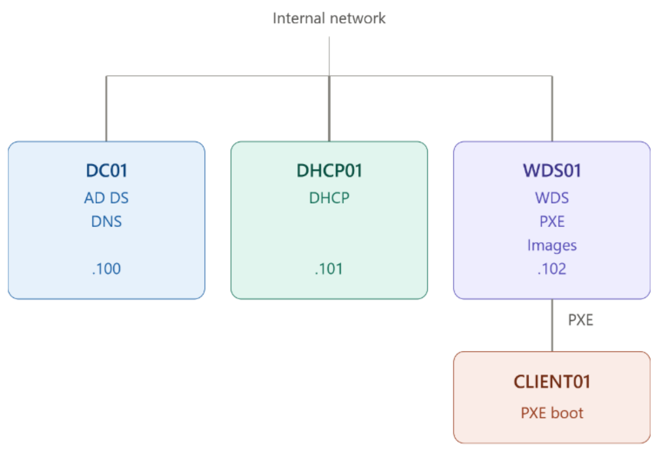
3. Infrastructure Overview

The organization uses separate servers for DHCP and WDS.

Recommended Lab / Enterprise Architecture

Server	Role	Example IP
DC01	AD DS + DNS	`192.168.1.100`
DHCP01	DHCP Server	`192.168.1.101`
WDS01	Windows Deployment Services	`192.168.1.102`
CLIENT01	Deployment Target	DHCP

The servers communicate through the organization's internal network.



4. Understanding the Roles

4.1 Active Directory Domain Services

DC01 provides:

- Active Directory
- Domain authentication
- Computer accounts
- User accounts
- Group Policy
- Domain management

Example:

```
Domain: lab.local
DC01:   192.168.1.100
```

4.2 DNS

DNS allows computers to locate domain services.

Example:

```
lab.local
```

The WDS server and clients should use the organization's internal DNS server.

Example:

```
Preferred DNS:
192.168.1.100
```

Do not configure domain computers to use public DNS such as:

```
8.8.8.8
1.1.1.1
```

as their primary DNS when they need to communicate with Active Directory.

5. DHCP Server

DHCP01 is responsible for providing network configuration to clients.

DHCP provides:

- IP address
- Subnet mask
- Default gateway
- DNS server
- DNS domain information

Example:

```
DHCP01
IP: 192.168.1.101
```

Example client lease:

```
IP Address:      192.168.1.150
Subnet Mask:     255.255.255.0
Gateway:         192.168.1.1
DNS:             192.168.1.100
DNS Domain:      lab.local
```

6. WDS Server

WDS01 provides Windows deployment services.

Example:

```
WDS01
IP Address: 192.168.1.102
```

WDS is responsible for:

- PXE boot
- Windows PE boot files
- Windows installation images
- Deployment of Windows computers
- Optional custom/golden images

7. Prerequisites

Before installing WDS, verify the following.

Server Requirements

The WDS server should have:

- Windows Server 2019/2022 or supported version
- Static IP address
- Working DNS
- Network connectivity
- Sufficient storage
- Domain membership when using AD-integrated WDS
- NTFS storage for deployment files

Example:

```
Computer Name: WDS01
IP Address:    192.168.1.102
DNS:           192.168.1.100
Domain:        lab.local
```

8. Configure the WDS Server Network

Configure a static IP.

Example:

```
IP Address:      192.168.1.102
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.1.1
Preferred DNS:   192.168.1.100
```

Verify:

```
ipconfig /all
```

Test DNS:

```
nslookup lab.local
```

Test connectivity to DC01:

```
ping 192.168.1.100
```

Test the domain:

```
nslookup dc01.lab.local
```

9. Join WDS01 to the Domain

Rename the server:

```
WDS01
```

Join:

```
lab.local
```

Restart the server.

Verify:

```
systeminfo | findstr /B /C:"Domain"
```

Expected:

```
Domain: lab.local
```

10. Install WDS

Open:

```
Server Manager
```

Select:

```
Manage → Add Roles and Features
```

Choose:

```
Role-based or feature-based installation
```

Select:

```
WDS01
```

Select:

```
Windows Deployment Services
```

Install:

- Deployment Server
- Transport Server

Complete the installation.

11. Create the RemoteInstall Folder

Create a dedicated deployment volume when possible.

Example:

```
D:\RemoteInstall
```

Avoid placing large deployment images on the operating system volume when the server has a dedicated data disk.

Example:

```
C:\
 └── Windows Server

D:\
 └── RemoteInstall
```

12. Configure WDS

Open:

```
Server Manager → Tools → Windows Deployment Services
```

Expand:

```
Servers
└── WDS01
```

Right-click:

```
WDS01 → Configure Server
```

Select:

```
Integrated with Active Directory
```

Confirm:

```
lab.local
```

Set the Remote Installation Folder:

```
D:\RemoteInstall
```

13. Configure PXE Response

For a controlled lab environment, configure WDS to:

```
Respond to all client computers (known and unknown)
```

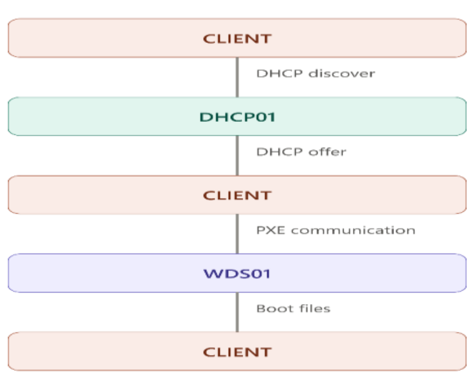
This allows newly installed computers to PXE boot without manually creating a prestaged computer account first.

> **Important:** In a production environment, consider restricting PXE responses according to the organization's security policy.

14. DHCP and WDS Integration

> **Important:** Because DHCP01 and WDS01 are separate servers, do not configure WDS as if DHCP is installed locally.

The basic process is:



The DHCP server provides the client's IP configuration.

WDS provides the deployment boot environment.

15. DHCP Configuration for PXE

The exact PXE configuration depends on the network architecture, firmware type, and WDS environment.

Important: Avoid blindly configuring old-style DHCP options 66 and 67.

For modern environments, PXE forwarding/IP helper configuration is generally preferred, particularly when DHCP and WDS are on different subnets.

For a simple same-subnet lab:

```
DHCP01
192.168.1.101

WDS01
192.168.1.102

CLIENT
DHCP assigned
```

All three should be able to communicate on the same deployment network.

16. Network / VLAN Considerations

If the client and WDS server are on different VLANs/subnets, configure the network infrastructure to forward DHCP/PXE traffic appropriately.

Example:

```
VLAN 10
Servers
192.168.1.0/24

VLAN 20
Workstations
192.168.20.0/24
```

The network infrastructure must allow the workstation VLAN to reach:

```
DHCP01
WDS01
DNS/DC
```

An IT Network Administrator may need to configure DHCP relay/IP helper addresses.

17. Add the Boot Image

WDS requires a boot image.

A standard Windows installation media normally contains:

```
sources\boot.wim
```

In WDS:

```
WDS01
└── Boot Images
```

Right-click:

```
Boot Images → Add Boot Image
```

Select:

```
sources\boot.wim
```

Complete the wizard.

The image should appear under:

```
Boot Images
└── Microsoft Windows Setup
```

18. Add the Install Image

WDS also requires an operating-system installation image.

Normally:

```
sources\install.wim
```

If the Windows ISO contains:

```
install.esd
```

instead of:

```
install.wim
```

convert/export the required edition to WIM.

Example:

```
dism /Get-WimInfo /WimFile:E:\sources\install.esd
```

Identify the correct Windows edition.

Example:

```
dism /Export-Image /SourceImageFile:E:\sources\install.esd /SourceIndex:6
/DestinationImageFile:C:\install.wim /Compress:max /CheckIntegrity
```

Then add:

```
C:\install.wim
```

to WDS.

19. Create an Install Image Group

In WDS:

```
Install Images
```

Right-click:

```
Add Install Image
```

Create a group such as:

```
Windows 10
```

or:

```
Windows 11
```

Select the appropriate install.wim.

Example:

```
Install Images
└── Windows 10
    └── Windows 10 Pro
```

20. Start the WDS Service

Check the service:

```
Get-Service WDSServer
```

Start it if necessary:

```
Start-Service WDSServer
```

Verify:

```
Get-Service WDSServer
```

Expected:

```
Status : Running
```

21. Verify DHCP

On DHCP01:

```
Get-Service DHCPServer
```

Expected:

```
Status : Running
```

Verify that the DHCP scope is active.

Example:

```
Scope:
192.168.1.0/24

Available addresses:
192.168.1.110 - 192.168.1.200
```

Important: Do not place static server addresses inside the dynamic range.

Example:

```
192.168.1.100  DC01
192.168.1.101  DHCP01
192.168.1.102  WDS01

192.168.1.110+ DHCP clients
```

22. PXE Boot Test

Configure the target computer to boot from the network.

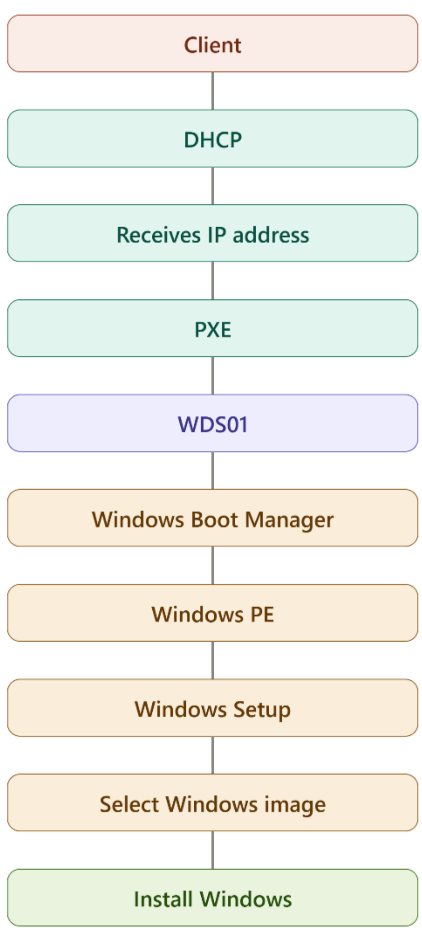
For a VM, ensure its network adapter is connected to the same virtual network as the deployment servers.

Start the client.

Select:

```
Network/PXE Boot
```

The expected sequence is:



23. Successful PXE Boot

A successful deployment should eventually display:

```
Windows Deployment Services
```

followed by the Windows installation environment.

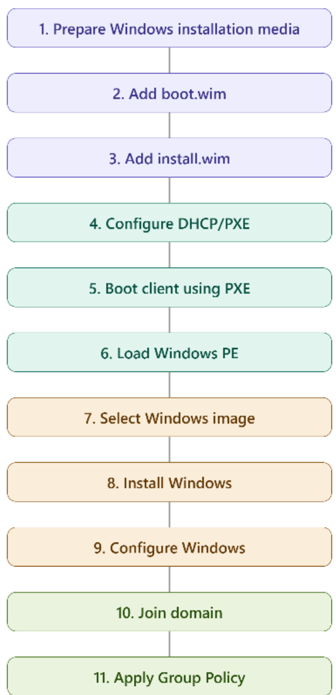
The technician should be able to select the available Windows image.

Example:

```
Windows 10 Pro
```

24. WDS Deployment Workflow

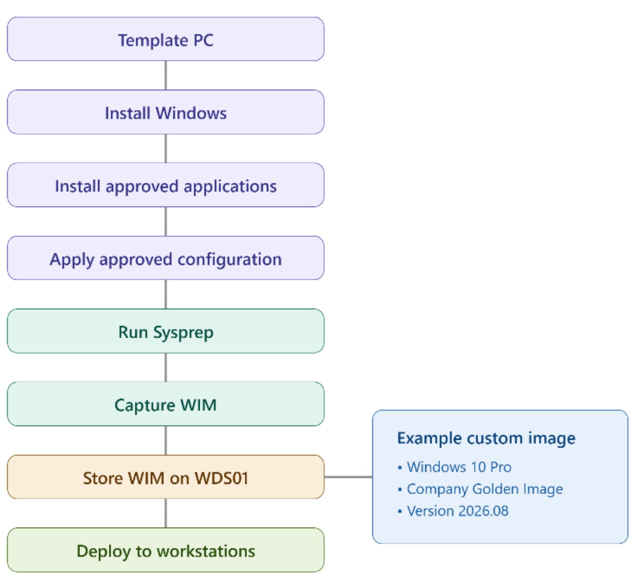
The standard deployment process is:



25. Golden Image Deployment

For organizations deploying many identical workstations, IT may maintain a standardized Windows image.

Example:



Example custom image:

```
Windows 10 Pro
Company Golden Image
Version 2026.08
```

The image should be version-controlled and documented.

26. Recommended Image Naming

Avoid vague names such as:

```
image.wim
new.wim
test.wim
final.wim
```

Use descriptive names:

```
Win10Pro-Golden-2026-08.wim
Win11Pro-Golden-2026-08.wim
```

For example:

```
WDS01
└── Images
    ├── Win10Pro-Golden-2026-08.wim
    └── Win11Pro-Golden-2026-08.wim
```

27. Verification Checklist

Before declaring WDS operational, IT should verify:

- ☐ WDS01 has a static IP.
- ☐ WDS01 can resolve the AD domain.
- ☐ WDS01 can communicate with DC01.
- ☐ WDS01 is joined to the domain.
- ☐ WDS role is installed.
- ☐ RemoteInstall folder exists.
- ☐ WDS service is running.
- ☐ DHCP01 is operational.
- ☐ DHCP scope is active.
- ☐ DHCP clients receive valid IP addresses.
- ☐ Client can reach WDS01.
- ☐ Boot image exists.
- ☐ Install image exists.
- ☐ PXE boot is enabled.
- ☐ Client successfully reaches Windows Deployment Services.
- ☐ Windows image appears.
- ☐ Windows installation completes successfully.

28. Basic Troubleshooting

Problem: Client does not receive an IP address

Check:

```
Get-Service DHCPServer
```

Verify:

- DHCP scope is active.
- Client is connected to the correct network.
- DHCP traffic is not blocked.
- Virtual network configuration is correct.

Problem: Client receives an IP but PXE does not start

Check:

- WDS service
- Network connectivity
- PXE configuration
- Firewall
- Client firmware mode

- DHCP/PXE forwarding

On WDS01:

```
Get-Service WDSServer
```

Problem: WDS starts but no Windows image appears

Check:

```
WDS01
├── Boot Images
│   └── boot.wim
│
└── Install Images
    └── Windows image
```

Verify that the WIM files are valid.

Problem: install.wim is missing

The Windows media may contain:

```
install.esd
```

instead.

Check:

```
dism /Get-WimInfo /WimFile:E:\sources\install.esd
```

Export the required edition to WIM.

Problem: PXE works before Sysprep but not afterward

> **Important:** Do not immediately assume Sysprep is the cause.

Check:

1. Client network adapter.
2. Virtual network.
3. DHCP lease.
4. WDS service.
5. WDS boot files.
6. Firmware mode.
7. Boot order.
8. WDS server connectivity.

Test PXE with a normal client first before troubleshooting the captured image.

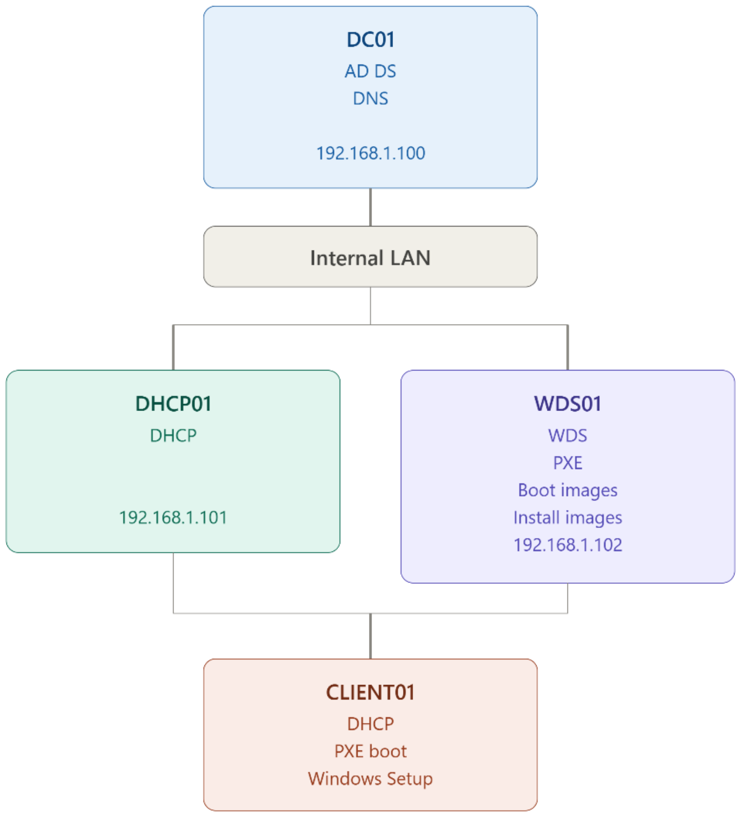
29. Important Separation of Responsibilities

For this organization's infrastructure:

Service	Server	Responsibility
AD DS	DC01	Domain authentication
DNS	DC01	Name resolution
DHCP	DHCP01	IP address assignment
WDS	WDS01	Windows deployment
File Server	FILESRV01	File storage
Print Server	PRINT01	Printer management

This separation makes the environment easier to manage, troubleshoot, and expand.

30. Final Recommended Architecture



Key Rule

DHCP assigns the IP address. WDS provides the Windows deployment environment.

They are separate services and can—and in your lab, will—run on separate servers.

The most important thing is that DHCP/PXE traffic can reach WDS01, and that WDS01 has the correct boot.wim and installation image configured.